Check Point Lab Setup and Administration

CCSA101 Introduction to Quantum Security

CCSA101-LAB01 ● Consolidated practical laboratory ● Individual work

Estimated time: 6 to 8 hours excluding downloads Maximum marks: 100 Due date: as displayed in the LMS

Build your own A-SMS management lab, explain the Check Point architecture, and explore Gaia and SmartConsole. Complete Parts A, B and C in sequence and submit one report. The entire activity is assessed once out of 100 marks.

Use only the supplied A-SMS virtual machine. Install SmartConsole on your existing physical Windows computer and access Gaia through a browser and the VMware console. Other devices may appear as preconfigured objects; they are observed only and do not need to be deployed or powered on.

Assessment structure

Part	Work required	Marks
A	Set up A-SMS and establish management access	40
B	Explain the architecture and administration decisions	25
C	Explore Gaia and navigate SmartConsole	35
Total	One activity and one PDF submission	100

Learning outcomes

- Prepare VMware, a private management network and a working A-SMS interface.
- Authenticate to the correct management server using SmartConsole.
- Explain management, gateway enforcement and the role of administration tools.
- Collect and interpret read-only system, policy, object and log observations.

Before you begin

You need Windows administrator access, hardware virtualization enabled, the complete instructor-supplied A-SMS package, the R82 SmartConsole installer and the supplied Gaia, SmartConsole and Expert credentials. Keep all VMX and VMDK files together. Use the course slides and official references listed at the end.

For A-SMS, the course guide specifies 12 GB RAM recommended, 8 GB minimum, 8 CPU cores and a 500 GB virtual disk. Preserve the supplied disk. A classroom allocation such as 4 vCPU needs instructor agreement. Leave sufficient physical RAM for Windows and SmartConsole. The LMS displays the deadline and submission window.

Part A Prepare VMware and the private network

Step 1 Install the Windows tools

Download and install VMware Workstation Pro for Windows using the official Broadcom instructions in the references. Restart Windows if requested. Obtain the R82 SmartConsole desktop installer from the official Check Point page or your instructor and install its prerequisites. Record both software versions and sources as evidence A1.

Step 2 Load A SMS and check its adapter

Extract A-SMS into a writable local folder. In VMware, select File > Open and choose A-SMS.vmx. Confirm its name, hardware and disk. Keep it powered off. Back up the VMX, close its VMware tab and open the VMX in Notepad. Add the following line once if absent, or correct the existing value:

```
ethernet0.virtualDev = "vmxnet3"
```

Save as .vmx, without .txt, then reopen it. In Network Adapter settings select Custom > VMnet8 and Connect at power on. Keep one adapter. Capture the hardware summary, connection choice and VMXNET3 entry as A2.

Step 3 Set the VMnet8 network

Open Edit > Virtual Network Editor and select Change Settings if needed. Select VMnet8, choose NAT, connect the host virtual adapter and enable the local DHCP service. Apply this plan. Coordinate any change to an existing VMnet8 used by another lab.

Setting	Required value
VMnet8 subnet and mask	10.1.1.0 / 255.255.255.0
NAT Settings gateway	10.1.1.250
DHCP Settings pool	10.1.1.210 through 10.1.1.240
Windows VMnet8 adapter	10.1.1.201/24 with no default gateway
A-SMS eth0	Static 10.1.1.101/24 with state on

Capture the subnet, NAT gateway and DHCP pool dialogs as A3. DHCP remains enabled for automatic clients; the two static addresses above remain outside its pool. The lab subnet stays fixed when Wi-Fi changes. Leave physical Wi-Fi on its normal settings.

Part A Set the addresses and verify Gaia

Step 4 Set the Windows VMnet8 address

Open Windows Command Prompt as administrator. Run this as one command; each caret continues the command onto the next line. Apply it only to the named VMware adapter.

```
netsh interface ipv4 set address ^
  name="VMware Network Adapter VMnet8" ^
  source=static address=10.1.1.201 ^
  mask=255.255.255.0 gateway=none

ipconfig
```

Confirm VMnet8 shows 10.1.1.201 and mask 255.255.255.0. Capture A4. Keep your physical Wi-Fi and Ethernet settings unchanged.

Step 5 Boot A SMS and inspect eth0

Power on A-SMS and wait for the Gaia login prompt and service startup. Log in through the VMware console with the supplied Gaia account, normally admin. In Gaia Clish run show interface eth0. Apply the set commands below only when the displayed address or state needs correction:

```
show interface eth0

set interface eth0 ipv4-address 10.1.1.101 mask-length 24
set interface eth0 state on
save config
show interface eth0
```

Capture the actual IPv4 address, state and link status as A5. If only lo appears, shut down cleanly and recheck the powered-off VMXNET3 setting. Use ipv4-address with the letter v and Tab completion when needed.

Step 6 Check access from Windows

```
ping 10.1.1.101
```

Record the ping result, then open https://10.1.1.101 in your browser and confirm the Gaia login page loads. For a certificate warning, verify the address belongs to your own lab VM before proceeding. Capture these checks as part of A6. Test the applications even if ping is unanswered.

Keep the supplied guest routes and DNS unchanged. Local management access is the requirement here. Internet access from the VM is a separate routing and DNS test and is not required for completion.

Part A Connect SmartConsole

Step 7 Verify the server and sign in

Launch the R82 SmartConsole client on Windows. Enter 10.1.1.101 as the management server and use the supplied SmartConsole account, normally cpadmin. The Gaia and SmartConsole accounts serve different interfaces.

When the first connection displays an ICA fingerprint, retrieve the fingerprint through your trusted A-SMS VMware console. Enter Expert mode using the supplied credentials, run the command and return to Clish:

```
expert
cp_conf finger get
exit
```

Compare the locally retrieved value with the SmartConsole prompt before trusting the connection. A mismatch must be investigated with your instructor. If the supplied Expert credentials are unavailable, record the limitation and obtain a trusted fingerprint from the instructor; do not reset credentials.

After authentication, open Gateways & Servers and select A-SMS. Record the displayed IP, version, status and license status. Capture an authenticated session and explain your fingerprint check as A6.

Checkpoint for Part A

Your own A-SMS must be reachable in Gaia and through an authenticated SmartConsole session. A successful ping alone is insufficient. If blocked, retain genuine partial evidence, the last successful step and the exact failure for feedback.

Troubleshoot the first failing check

Observation	Next check
VM cannot boot	Virtualization, free host RAM and matching VMX and VMDK files.
eth0 is missing	Power off and verify the VMXNET3 key in the correct VMX file.
eth0 link is down	VMware connection, Custom VMnet8 and Gaia interface state.
Gaia works but SmartConsole fails	R82 client, server IP, correct account and management startup.
Other objects are unavailable	Their VMs may be off; record the state without changing them.

Use the VMware console for Gaia Clish and Expert commands. SmartConsole's Command Line panel is a separate management interface. Record symptoms, checks, corrections and retest results in your issue log.

Part B Explain the Check Point architecture

Answer all five questions in your own words. Each question carries 5 marks, for a total of 25. Aim for 700 to 1,000 words across all answers, excluding diagram labels and references. Cite relevant course slides or official documentation. The gateway in the diagram is conceptual; do not deploy it.

B1 Explain the architecture

Draw a labelled three-tier diagram showing SmartConsole on Windows, the Security Management Server and a Security Gateway. Explain each role, where policy and object definitions are maintained, and the difference between changing a policy and enforcing it.

Marks: labelled diagram 2; accurate roles and policy flow 3.

B2 Explain management services and deployment

Explain three Security Management Server functions. Compare an on-premises management deployment with Smart-1 Cloud, then recommend one for an organisation that wants less on-premises hardware maintenance. State one tradeoff.

Marks: three explained functions 3; comparison and justified recommendation 2.

B3 Choose the correct administration interface

Choose and justify the tool for each task: inspect eth0 and its subnet mask; review an Access Control rule; inspect operating-system resource use; view security logs. Then distinguish Gaia Clish, Expert mode and SmartConsole's Command Line panel.

Marks: four appropriate tool choices with reasons 3; correct command-context distinction 2.

B4 Explain security features and policy handling

Choose two Security Gateway Software Blades or features covered in the course. Give a realistic use case for each. Explain why publishing a management session and installing policy on a gateway are separate actions.

Marks: features and relevant use cases 2; accurate publishing and installation explanation 3.

B5 Interpret your own lab evidence

Explain why the private VMnet8 addresses can remain fixed when Wi-Fi changes and why enabled VMware DHCP does not alter the static A-SMS address. Explain what ping proves compared with SmartConsole login. Finally, explain why an unavailable gateway object or empty log view needs further evidence before you diagnose a fault.

Marks: private networking and DHCP 2; connectivity evidence 1; reasoned interpretation of status and logs 2.

Part C Explore Gaia Portal

Begin after Part A. Use only A-SMS. This part is read-only: inspect existing values and close property windows with Cancel. Keep routes, DNS, accounts, licenses, Software Blades and policies unchanged. Record your actual observations, even when they differ from an example.

C1 Inspect the system overview

Log in at https://10.1.1.101 with your Gaia account. Record hostname, product release, build information where displayed, uptime, CPU use and memory use. Explain briefly why resource readings can change. Capture G1.

C2 Compare the navigation modes

Compare Basic and Advanced views where available. Record three navigation areas and their purposes, and identify an option available only in Advanced view. Capture both views as G2. If labels or permissions differ, record the actual interface and limitation.

C3 Inspect the management interface

Open Network Management > Network Interfaces and locate eth0. Record its IPv4 address, subnet mask, administrative state, link status and MAC address where shown. Compare the address with Part A and the Clish result. Close any Edit dialog with Cancel. Capture G3; A5 can be cross-referenced for the Clish comparison.

C4 Inspect routes and DNS

Locate the IPv4 routing or static-route page and the DNS configuration page. Record the default gateway if present and the configured DNS servers. Explain the difference between a default gateway and a DNS resolver. Capture both views as G4 and state whether VM internet access was actually tested.

The supplied image may retain the course default route through 10.1.1.1, while the VMware NAT gateway is 10.1.1.250. Record the actual value without changing it. Local SmartConsole access does not establish that the VM can reach the internet.

C5 Locate maintenance and administration functions

Locate User Management, time or NTP settings, and the Maintenance backup or snapshot pages. Record each actual menu path and purpose. Explain the difference between a configuration backup and a system snapshot. Capture one relevant maintenance page as G5. Inspect the controls without running a backup, snapshot, update or restore.

Record observations consistently

For every exploration task, record its code, menu path or command, actual observation, a short interpretation and the matching figure ID. Keep observed facts separate from an explanation or hypothesis. If an expected page is unavailable, show the result and explain the limitation.

Part C Inspect Gaia Clish and Expert mode

C6 Collect system information in Clish

In the A-SMS VMware console, log in to Gaia Clish and run these read-only commands. Use Tab completion if needed. Capture the outputs as G6 and explain what each reveals. Reuse Part A evidence when it already shows the same current value clearly.

```
show version all
show uptime
show interfaces all
show interface eth0
```

Compare the release and eth0 address with Gaia Portal. Note any difference in how the two interfaces present the same information.

C7 Recognise the Expert shell

With the supplied Expert credentials, enter Expert mode, record the changed prompt and run only these inspection commands. Capture G7, then return to Clish:

```
expert
whoami
pwd
exit
```

Explain how the prompt and command environment differ from Clish. If Expert access is unavailable, record the result and explain the expected distinction without resetting a password. Your fingerprint check in Part A can be cross-referenced but does not replace the shell observations here.

C8 Identify the command environment

Locate SmartConsole's Command Line panel later in the client. Viewing it is sufficient; no scripts or management commands are required. Explain why ip route or ping -c may be rejected there and why Linux diagnostics belong in the appropriate Expert session.

A short issue log

For each issue, record the symptom, the command or page used to investigate it, any correction made during Part A, and the retest result. In Part C, document unexpected findings without making configuration changes. If no correction was needed, state that explicitly.

Example structure: task code | symptom | check | observed result | correction or next step | figure ID. Describe your own results rather than copying an expected output.

Part C Navigate SmartConsole

Continue in the authenticated R82 SmartConsole session connected to 10.1.1.101. Observe the existing environment. Close object properties with Cancel and leave the session unchanged. Do not publish changes or install policy during this exploration.

C9 Inspect Gateways and Servers

Select A-SMS and inspect Summary, Alerts and Licenses where accessible. Record its address, version, device status, Software Blades and license status, including any exact alert text. Capture S1. Other preconfigured devices may be unavailable because only A-SMS is powered on; record that limitation.

C10 Inspect an existing policy

Open Security Policies and locate the supplied Access Control policy, such as Alpha-Standard if present. Record its actual name. Inspect one existing rule and describe its Name, Source, Destination, Services and Applications, Action and Track fields. Explain its intended behaviour from the visible values. Capture S2.

C11 Explore objects and administrative controls

Use the Objects pane or Object Explorer to inspect A-SMS and one other existing object. Record each type, relevant address or identifier, and one property that explains its purpose. Locate Manage & Settings, session controls and the Tasks window and record their purposes. Capture the relevant views as S3.

C12 Inspect Logs and Events

Record the selected log source or server and the time range. If a log entry is available, report its time, origin, action and one meaningful detail. If no results appear or the log source is unavailable, capture the exact state and explain why it does not prove that no traffic or security events occurred. Capture S4.

A separate Log Server may be present as an object but is not deployed in this activity. Do not create traffic, power on another VM or modify logging settings to force a successful screenshot.

C13 Locate publication and installation controls

Identify the Publish and Install Policy controls and describe their purposes. Observe their location without using them. Cross-reference your answer to B4. Locate the Command Line panel for C8 and explain how it differs from the two Gaia command environments.

If your account cannot access a requested policy, object property or page, document the visible result and explain what remains unverified. Genuine limitations are valid observations; invented successful results are not.

Submit one consolidated report

Submit one PDF named CCSA101_LAB01_StudentID.pdf through the single LMS activity. Replace StudentID with your actual registration number and use underscores in place of slashes. Put your name, student ID and completion date on page 1.

Arrange the report as Part A Setup Evidence, Part B Theory Answers, Part C Observation Record, then Issue Log and References. All three parts belong in this one submission. A practical target is 12 to 18 pages including screenshots; readability and complete evidence matter more than length.

Evidence checklist

ID	Required evidence
A1	VMware and R82 SmartConsole versions and download sources.
A2	A-SMS hardware, Custom VMnet8 connection and VMXNET3 entry.
A3	VMnet8 NAT subnet, NAT gateway and DHCP pool settings.
A4	Windows ipconfig showing VMnet8 at 10.1.1.201/24.
A5	Gaia eth0 address, administrative state and link status.
A6	Windows ping, Gaia access, authenticated SmartConsole and fingerprint method.
G1 to G5	Gaia overview, views, interface, routes and DNS, plus maintenance observation.
G6 and G7	Clish outputs and Expert prompt with whoami and pwd, or documented limitation.
S1 to S4	SmartConsole status, policy rule, objects and controls, plus logs view.

Use screenshots from your own lab. Caption each figure with its ID, task and what it proves. One clear figure may cover several related tasks, and evidence may be cross-referenced instead of repeated. Keep passwords and complete license keys out of screenshots.

Written content and authenticity

Include all five Part B answers, the labelled architecture diagram, the Part C observation record and your issue log. Cite the course slide numbers or official pages used. Disclose any assistance in line with course policy. Your screenshots, observations and explanations must represent your own work.

If blocked, submit genuine partial work and the last successful checkpoint, exact failure and diagnostic checks. Record unavailable features and permission limits explicitly.

Marking rubric and learning resources

This is one 100-mark assessment. Part A contributes 40 marks, Part B 25 marks and Part C 35 marks. Part B submarks appear with each question. The rubric below governs the practical evidence.

Criterion	Marks
A Software versions and sources recorded	5
A A-SMS hardware and VMXNET3 adapter preparation	8
A VMnet8 NAT plan and Windows host address	12
A Gaia eth0 configuration and verification	8
A Authenticated SmartConsole and server identity check	7
B Five theory answers at 5 marks each	25
C Gaia Portal observations C1 to C5 with evidence	12
C Clish and Expert observations C6 to C8	8
C SmartConsole observations C9 to C13	10
C Clear observation record and issue log with readable original evidence	5
Total	100

Full marks require accurate work supported by readable evidence and clear interpretation. Partial marks apply to correct but incomplete work. An unavailable feature can earn interpretation marks when you document the actual result and explain its limits; it cannot establish an unverified technical outcome.

Course alignment

This consolidated activity combines management lab preparation, course architecture theory, and the Explore Gaia and Connect to and Navigate SmartConsole practical topics. In the supplied 65-slide CCSA101 deck, the practical topics appear on slides 64 and 65. Only A-SMS is required.

Learning resources

CCSA101 course slides and CCSA R82 V1.0 Lab Setup Guide

Broadcom VMware Workstation download instructions

Check Point R82 SmartConsole download guidance

Check Point R82 Gaia Administration Guide

Check Point R82 ICA fingerprint verification

Check Point R82 Security Management Administration Guide